

Aasma Aas *Data Engineer*

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📍 Lahore, Pakistan

🐙 GitHub

🔗 Aasma_Portfolio

🌐 LinkedIn

Professional Experience

2025/01 – present
Lahore, Pakistan

Data Engineer

DatumLabs 🌐

Responsibilities:

- Designed and implemented scalable **data pipelines** using DBT, ClickHouse, Snowflake, and AWS MWAA (Airflow).
- Built and optimized **data models** in ClickHouse and Snowflake to support high-performance analytics.
- Developed interactive **dashboards in Apache Superset** for data visualization and stakeholder reporting.
- Mentored and trained junior team members on DBT, ClickHouse, and Superset best practices.
- Ensured **data quality, reliability, and performance** through testing, reviews, and workflow optimization.

2023/12 – 2025/01
Lahore, Pakistan

Associate Data Engineer

DotLabs 🌐

- Designed and executed scalable ETL processes on **AWS** using EC2, AWS Lambda, and AWS Glue, incorporating Python-based automation scripts.
- Built and optimized ETL pipelines using DBT, Airbyte, Azure, and AWS to enable seamless and accurate data movement and transformation across various platforms.
- Utilized Airbyte and Airflow for workflow orchestration, automating complex ETL tasks efficiently.
- Developed **CI/CD pipelines** to streamline deployment processes and improve development efficiency.
- Integrated and automated data workflows for HubSpot and GA4 APIs, facilitating smooth ingestion and analysis of marketing and customer behavior metrics.
- Created the interactive dashboards in QuickSight and Power BI.

2021/06 – 2023/06
Lahore, Pakistan

Front-End Developer

Arhamsoft Pvt Ltd 🌐

- Lead projects, mentor junior team members, and work closely with designers to ensure the correct UI components, layouts, and visual implementation.
- Designed an animated website using GSAP library in reactJs or simple javascript.
- Implemented various features and functionality using javascript, reactjs, and jQuery.

2020/12 – 2021/06
Lahore, Pakistan

Front-End Developer

ITD

Education

2022 – 2024
Lahore, Pakistan

MPhil Computer Science

University of Management and Technology 🌐

Projects

Thesis: Suspicious Activity Detection [↗](#)

CNN+LSTM | CNN+VGG16 | Keras | transfer learning model (all type of DenseNet, Resnet and VGG16+19) | NumPy, Python, Flask, OpenCV, Keres, TensorFlow

The study uses a CNN-LSTM model for real-time crime activity detection in video surveillance footage. The CNN is used for feature extraction, while the LSTM captures temporal dependencies.

Sojo Industries | Associated with DatumLabs [↗](#)

AWS MWAA | DBT | Snowflake | OpenSearch

Designed and implemented end-to-end data pipelines on AWS MWAA (Airflow) using DBT and Snowflake, with data ingestion from MySQL and PostgreSQL. Built destination mapping on OpenSearch to enable efficient search and analytics, and introduced Cosmos for DBT within AWS MWAA to streamline model execution and orchestration.

Markets4You | Associated with DatumLabs [↗](#)

ClickHouse | DBT | Superset | Docker

I worked on designing and improving the ClickHouse database to ensure it could handle fast and efficient data analysis for real-time trading insights. I built DBT pipelines for ClickHouse, facilitating efficient incremental, full data loading and transformation. Additionally, I created dashboards using Apache Superset.

Hospi Housing | Associated with Dotlabs

AWS Dynamo DB | Kinesis | S3 | Glue | Athena | QuickSight

Executed an ETL pipeline and business intelligence processes on AWS. The pipeline extracts live data from DynamoDB via Kinesis, transforms it using Glue scripts, and crawls into AWS Athena. It then integrates with AWS QuickSight to design dashboards that provide stakeholders with data insights.

Brainivoy ETL Pipeline | Associated with Dotlabs

Hubspot | Python Flask | Airbyte | AWS S3 | AWS Glue

Designed and implemented an ETL pipeline from HubSpot to AWS. Utilizing the HubSpot developer portal, we built a Flask application that uses OAuth 2.0 for user authentication, extracted data via Airbyte configured on AWS EC2, and created a data lake on AWS using S3 and Glue jobs.

Azure Meal | Associated with Dotlabs

Azure Function App | Azure SQL Server Database | Power BI

Designed an ETL pipeline on Azure, where data extraction is triggered by an Azure Function App in response to specific events. The processed data is stored in an Azure SQL Server Database, followed by the generation of a Power BI report.

Sindri Media | Associated with Dotlabs [↗](#)

Python | SQL | Airflow | DBT | S3 | EC2

Implemented optimized stored procedures and created indexes on frequently queried columns, resulting in significant performance improvements. Additionally, migrated the project to DBT (Data Build Tool) to streamline and enhance the efficiency of data pipelines.

Skills

Languages

Python

Cloud & Big data tools

AWS MWAA, Open Search(Elastic Search), Amazon EC2, S3, Athena, DynamoDB, Glue, IAM, QuickSight, PySpark, Airflow, DBT, PowerBi, Airbyte, Apache Superset, Docker

Front-End skills & Web Framework

Python Flask, React.js, Html, React Bootstrap, GSAP, Shopify Polaris, CSS, jQuery, Tailwind CSS

Databases

Clickhouse, Snowflake, PostgreSQL, MYSQL

Publications

2025/03

Efficient and Sustainable Video Surveillance Using CNN-LSTM Model for Suspicious Activity Detection

A. Aas (**Corresponding Author**), H. Naveed, J. Asghar, Z. Khanum, S. Khaleel, and T. Noreen, "Efficient and Sustainable Video Surveillance Using CNN-LSTM Model for Suspicious Activity Detection," VFAST Transactions on Software Engineering (VTSE), vol. 13, no. 1, pp. 60–71, Mar. 2025.

DOI: <https://doi.org/10.21015/vtse.v13i1.2023> .

Indexing: Scopus; HEC Pakistan – Y Category. ISSN(e): 2309-3978, ISSN(p): 2411-6246.

2024/12

A Blockchain-Based Framework to Make the Citrus Crop Supply Chain Transparent and Reliable in Agriculture

S. Kanwal, S. Nazir, S. M. A. Gilani, J. Asghar, and A. Aas, "A Blockchain-Based Framework to Make the Citrus Crop Supply Chain Transparent and Reliable in Agriculture," VAWKUM Transactions on Computer Sciences (VTCS), vol. 12, no. 2, pp. 205–220, Dec. 2024.

DOI: <https://doi.org/10.21015/vtcs.v12i2.1954> .

Indexing: Scopus; HEC Pakistan – Y Category. ISSN(e): 2308-8168, ISSN(p): 2411-6335.

Certificates

- Stanford WIDS Datathon Workshop 2023
- Deep Learning with Keras and Tensorflow 
- Machine Learning with Python 